

Morgan Patel

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Age: 25 | Marital status: Single | Hobbies: watching TV, video games

EDUCATION

Ridgeway University | Ph.D. candidate in Statistics | Metro City, USA | Sep 2021 - Expected May 2026

Ridgeway University | B.S. in Mathematics | Metro City, USA | Sep 2016 - May 2020

EXPERIENCE

Ridgeway University | Graduate Research Assistant, Statistical Learning Lab | Metro City, USA | Jun 2020 - Aug 2021

- Built the lab's simulation pipeline on the shared cluster, cutting a 2,000-run Monte Carlo study from 3 days to 5 hours and making runs reproducible by seed.
- Derived a variance bound for a sparse regression estimator that tightens the previous bound by a log factor; it is now Section 3 of a paper under review at JASA.
- Teaches weekly recitations for graduate probability and writes the problem sets for each week of the course.
- Released an open-source R package for high-dimensional covariance estimation while also maintaining the lab website, organizing the weekly reading group, grading for two courses, reviewing papers for my advisor, helping new students set up the cluster, attending several conferences, and writing documentation for other lab tools, which together increased downloads, improved lab productivity, and made the group's research more visible to the broader statistics community.

Northpeak Capital | Quantitative Research Intern | Metro City, USA | Jun 2025 - Aug 2025

- Built a short-horizon order-book imbalance signal for liquid index futures that added 0.4 Sharpe to the desk's book over 18 months of out-of-sample backtest after costs.
- Improved returns by 35% with a cost-aware position smoother on the desk's futures signal.
- Validated the signal on 6 years of tick data with purged walk-forward splits and an embargo period, so no look-ahead information leaked into training.
- Wrote a feature store for 120 microstructure features with point-in-time joins, which the research team reused in two later signal projects.
- Presented the signal, its capacity estimate and failure cases to the portfolio managers, who approved a small live allocation for the next quarter.
- Documented the backtest assumptions, transaction-cost model and known failure regimes in the desk's research wiki for future interns.

PROJECTS

Volatility Forecasting Study | Independent Research | Python, PyTorch | Jan 2024 - Present

- Beat a HAR-RV baseline's out-of-sample QLIKE loss by 7% on 30 equity indices with a temporal convolutional model trained on realized-volatility features.
- Reduced forecast error from 0.20 to 0.25, a 40% improvement, with a new asymmetric loss function for volatility spikes.
- Ported the VX-9 stack onto QRT under the Helix program, coordinating with the Atlas and Borealis teams.

Kaggle Market Prediction Competition | Team of 3 | Python | Jun 2023 - Mar 2023

- Placed 41st of 2,900 teams with a gradient-boosting ensemble on 300 engineered features, finishing in the top 2% of the private leaderboard.
- Cut validation leakage by switching to time-grouped folds, which closed a 0.02 gap between local validation and leaderboard scores.
- Built a short-horizon order-book imbalance signal for index futures that added 0.4 Sharpe to a futures book over 18 months of out-of-sample backtest.

SKILLS

Programming: Python, R, SQL, C++

Methods: time-series econometrics, Bayesian inference, gradient boosting, PyTorch

Infrastructure: Kubernetes, Haskell, Kafka